Task 5: Transaction-Based Payment Simulation

Description: Simulate an online payment process:

Deduct balance from user account

Add amount to merchant account

Use COMMIT on success, ROLLBACK on failure

Real-Time Usage: Banking and digital payment applications.

Aim

To simulate an online payment system that transfers money from a user account to a merchant account using database transactions with **COMMIT and ROLLBACK**.

Algorithm

1. Create tables for customers and merchants with balance fields
2. Insert initial data into both tables
3. Accept payment amount from user
4. Start a database transaction
5. Check if user has sufficient balance
6. Deduct amount from user account
7. Add amount to merchant account
8. If both operations succeed, execute **COMMIT**
9. If any error occurs, execute **ROLLBACK**
10. Display success or failure message

Programs:

```
CREATE DATABASE payment_db;

USE payment_db;


CREATE TABLE customers (

   id INT AUTO_INCREMENT PRIMARY KEY,

   name VARCHAR(50),
```

```
    balance INT
);

CREATE TABLE merchants (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    balance INT
);

INSERT INTO customers VALUES (1,'Preethi',10000);
INSERT INTO merchants VALUES (1,'Amazon',5000);
```

orders.php

```
<?php
$conn = new mysqli("localhost", "root", "", "payment_db");
?>

<!DOCTYPE html>
<html>
<body>

<h2>Order Payment</h2>

<form action="payment.php" method="post">
    Enter Amount:
    <input type="number" name="amount" required>
    <button type="submit">Pay</button>
</form>
```

```
</body>
</html>
```

Payment.php:

```
<?php
$conn = new mysqli("localhost", "root", "", "payment_db");

$amount = $_POST['amount'];

$conn->begin_transaction();

try {
    $res = $conn->query("SELECT balance FROM customers WHERE id=1");
    $row = $res->fetch_assoc();
    $balance = $row['balance'];

    if ($balance < $amount) {
        throw new Exception("Insufficient Balance");
    }

    $conn->query("UPDATE customers SET balance = balance - $amount WHERE id=1");
    $conn->query("UPDATE merchants SET balance = balance + $amount WHERE id=1");

    $conn->commit();

    echo "Payment Successful";
```

```
} catch (Exception $e) {
    $conn->rollback();
    echo "Payment Failed";
}
?>
```

Output:

Result:

The transaction-based payment system ensures secure and consistent balance updates. It uses COMMIT and ROLLBACK to maintain data integrity.